

Cross-Modal JEPA Gate-Blend Fusion for Lightweight RGB/Infrared Detection and Semantic Segmentation

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Abstract—This study evaluates RGB/infrared fusion for lightweight object detection and semantic segmentation, with particular attention to whether a learned cross-modal module improves over simple fusion. The main empirical finding is deliberately conservative: under the verified experimental protocol, pixel-level average fusion is the strongest overall representation. A dual-branch RGB/IR model with channel attention, spatial gating, and a cross-modal joint-embedding predictive architecture (CM-JEPA) objective is evaluated on 6025 aligned pairs from M3FD, MSRS, and RoadScene. YOLO11n is used for detection on M3FD, U-Net/MobileNetV2 for semantic segmentation on MSRS, and entropy, spatial frequency, average gradient, and normalized mutual information for image-level fusion diagnostics. The final proposed output is a gate-guided representation, $I_{A7} = G_{\uparrow} \odot I_{RGB} + (1 - G_{\uparrow}) \odot \text{rep}(I_{IR})$, because the raw decoder output could collapse visually. Average fusion achieved the best detection mAP@0.5 (0.7718), mAP@0.5:0.95 (0.5115), and segmentation mIoU (0.5551). CM-JEPA gate-blend was competitive for strict detection localization (mAP@0.5:0.95 = 0.5080), but weaker for segmentation (mIoU = 0.4774). The results support RGB/IR fusion for lightweight perception, but they also show that learned fusion complexity and image-level fusion metrics do not automatically imply superior downstream performance.

Index Terms—RGB-infrared fusion, joint-embedding predictive architecture, attention, YOLO11, U-Net, semantic segmentation, object detection, lightweight perception.

I. INTRODUCTION

Robust visual perception is central to intelligent transportation, surveillance, mobile robotics, and safety-oriented monitoring. Visible RGB cameras preserve color, texture, and fine edge structure, but their reliability decreases under low illumination, fog, rain, smoke, glare, and strong contrast changes. Infrared (IR) imaging captures thermal responses and can reveal pedestrians, vehicles, and lamps when visible imagery is degraded. However, IR imagery usually has weaker texture, limited color semantics, and sometimes coarse object boundaries. Infrared-visible image fusion (IVIF) therefore aims to combine complementary sensing properties into a representation that improves human inspection and downstream computer vision.

Recent IVIF work has produced increasingly sophisticated learned fusion models. DenseFuse introduced an autoencoder-based fusion paradigm [1]; task-aware methods such as SeA-Fusion and TarDAL connected fusion to segmentation and

detection [2], [3]; and CDDFuse and MRFS introduced decomposition, correlation, and mutually reinforcing task objectives [4], [5]. These studies motivate learned cross-modal fusion, but they also make a practical question sharper: when downstream models are deliberately lightweight and unchanged, is a learned fused image more useful than simple average fusion?

This project addresses that question through a controlled experimental study. A cross-modal attention model is implemented with dual encoders, SE-style channel attention [6], a spatial gate inspired by attention modules such as CBAM [7], and a JEPA-style latent prediction objective [8]. The method is compared with RGB-only, IR-only, and average fusion. Instead of relying only on classical image-quality metrics, the same fused representations are evaluated with YOLO11n detection [9] and U-Net/MobileNetV2 segmentation [10], [11].

The contributions are as follows:

- A complete real-data RGB/IR fusion pipeline was built over 6025 aligned pairs from M3FD, MSRS, and RoadScene.
- The study reports a negative but informative result: average fusion is the strongest overall baseline under the verified downstream protocol.
- CM-JEPA gate-blend is evaluated with the same fused image for image metrics, YOLO, and U-Net, correcting an earlier raw-decoder evaluation inconsistency.
- A0-A7 ablations are reported, while the downstream scope of A3-A6 is explicitly limited to avoid overgeneralizing from fusion-only metrics.
- The analysis connects failure modes to representation collapse, low-resolution gating, downstream task requirements, and the weak relationship between image-level fusion metrics and task performance.

II. RELATED WORK

A. Infrared-Visible Fusion

Classical methods such as Laplacian pyramid fusion combine multi-scale image coefficients using deterministic rules [12]. They are transparent and parameter-free, but they do not adapt to semantic task objectives. Deep IVIF methods learn fusion representations from data. DenseFuse uses a dense autoencoder to fuse visible and infrared features [1]. CDDFuse

Experimental pipeline and evaluation protocol

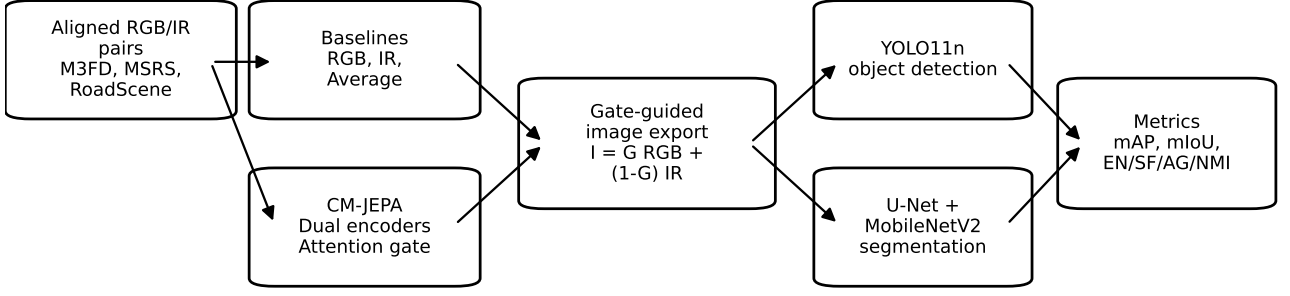


Fig. 1: End-to-end experimental pipeline. Paired RGB/IR data are standardized into manifests, processed by simple baselines and the proposed CM-JEPA gate-blend representation, and evaluated with both image-level fusion metrics and lightweight downstream perception tasks.

decomposes features into shared and modality-specific components and constrains the decomposition through correlation-driven learning [4]. These approaches are important references for learned fusion, but their image-quality improvements do not necessarily guarantee stronger object detection or semantic segmentation.

B. Task-Aware and Cross-Modal Perception

Task-aware methods directly optimize fusion for high-level vision. SeAFusion introduces semantic supervision into real-time IVIF [2]. TarDAL proposes target-aware dual adversarial learning and the M3FD benchmark for detection-oriented fusion [3]. MRFS jointly reinforces fusion and segmentation [5]. In RGB-X perception, CMX demonstrates the value of cross-modal interaction for semantic segmentation [13]. Compared with these larger or more task-coupled systems, the present work deliberately uses a modular fused-image adapter so that standard YOLO11n and U-Net/MobileNetV2 can remain unchanged. This design makes the comparison practical and reproducible, but it also limits how strongly the fusion model can be optimized for a particular downstream task.

C. JEPA-Style Representation Learning

Joint-embedding predictive architectures learn by predicting target representations from context representations rather than reconstructing pixels [8]. This is conceptually attractive for RGB/IR fusion because visible and thermal images measure different physical properties. A latent prediction objective can encourage shared scene structure without forcing one modality to reconstruct the other pixel-by-pixel. However, the present experiments show that JEPA-style representation learning is not sufficient by itself. The raw decoder path could collapse into a low-variance image, and the final valid representation had to use the learned gate as an image-level RGB/IR blend. The anti-collapse terms follow the general motivation of variance and covariance regularization [14], but the results indicate that task-aware supervision is still necessary for dense semantic preservation.

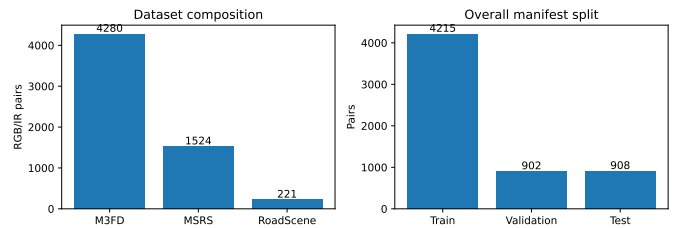


Fig. 2: Dataset and split statistics after standardization. The manifest contains 6025 RGB/IR pairs across M3FD, MSRS, and RoadScene.

III. MATERIALS AND METHODS

A. Datasets and Standardized Manifest

The verified manifest contains 6025 aligned RGB/IR pairs. M3FD contributes 4280 pairs and is used for object detection with six YOLO classes: people, car, bus, motorcycle, truck, and lamp. MSRS contributes 1524 standardized pairs, among which 1444 valid mask-bearing samples are used for segmentation. RoadScene contributes 221 paired samples for fusion-quality and qualitative analysis. The overall split contains 4215 training, 902 validation, and 908 test pairs. The M3FD YOLO split contains 2994 training, 641 validation, and 645 test images for each evaluated representation.

B. Baselines and Evaluation Scope

The downstream comparison includes RGB-only $I_{A0} = I_{\text{RGB}}$, IR-only $I_{A1} = \text{rep}(I_{\text{IR}})$, average fusion

$$I_{A2} = \frac{1}{2}I_{\text{RGB}} + \frac{1}{2}\text{rep}(I_{\text{IR}}), \quad (1)$$

and the final CM-JEPA gate-blend representation. Average fusion is intentionally retained as a strong low-complexity baseline. If a learned model cannot outperform it, the learned model must be justified through robustness, interpretability, or future task coupling rather than architectural novelty alone.

Learned SOTA methods such as DenseFuse and CDDFuse are discussed as contextual references rather than reported as same-protocol numeric baselines. They were not executed

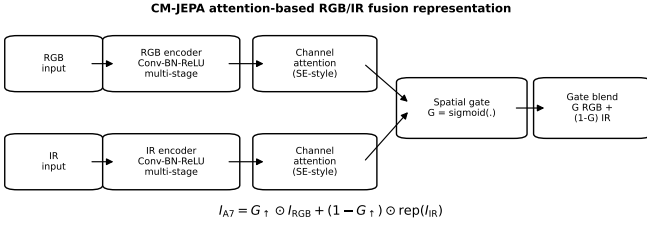


Fig. 3: Final proposed representation. The learned gate is used to blend RGB and replicated IR at image level; the raw decoder output is not used for final downstream evaluation.

through the verified YOLO/U-Net pipeline in this submission, and no DenseFuse or CDDFuse task metrics are claimed. This is an important boundary of the study: the results show that average fusion outperformed the implemented CM-JEPA gate-blend model under this protocol, not that average fusion is universally superior to all learned IVIF methods.

C. CM-JEPA Gate-Blend Fusion

Let F_R and F_I denote RGB and IR feature tensors from separate encoders. Channel attention produces F'_R and F'_I . A spatial gate is estimated as

$$G = \sigma(\phi([F'_R, F'_I])), \quad (2)$$

where $[\cdot, \cdot]$ denotes concatenation, ϕ is a lightweight convolutional gate predictor, and σ is a sigmoid. The latent fusion is

$$F_{ctx} = G \odot F'_R + (1 - G) \odot F'_I. \quad (3)$$

Although a feature adapter decodes F_{ctx} into a 3-channel image during training, the raw decoder output was not reliable as the final fused representation. The evaluated proposed image is therefore

$$I_{A7} = G_{\uparrow} \odot I_{RGB} + (1 - G_{\uparrow}) \odot \text{rep}(I_{IR}), \quad (4)$$

where $G_{\uparrow}$ is the spatial gate upsampled to image resolution. This same representation is used for fusion metrics, YOLO, and U-Net.

D. Objectives and Training

The fusion objective combines intensity and gradient preservation:

$$\mathcal{L}_{fus} = \|I_f - I_{ref}\|_1 + \lambda_g \|\nabla I_f - \nabla I_{ref}\|_1, \quad (5)$$

with $\lambda_g = 5.0$. The JEPA component predicts target latent tokens from context tokens:

$$\mathcal{L}_{jepa} = \|p(z_c) - \text{sg}(z_t)\|_2^2, \quad (6)$$

where $p(\cdot)$ is the predictor and sg denotes stop-gradient. Variance and covariance terms with weights 0.05 and 0.005 are used to discourage collapsed latent representations. The final configuration uses AdamW with learning rate 2×10^{-4} , weight decay 10^{-4} , EMA momentum 0.996, mask ratio 0.35, base channels 32, latent channels 512, attention reduction 16, and token dimension 256.

TABLE I: Implementation cost information saved by the notebook. Controlled latency was not benchmarked, so no runtime claim is made.

Component	Total params	Trainable params
Fusion image model	10.90M	10.90M
CM-JEPA wrapper	15.99M	12.75M

Checkpoint size for : 61.1 MB. Fusion tensors are $512 \times 32 \times 32$ for 512×512 inputs.



Fig. 4: Saved CM-JEPA training history from the reproducibility notebook. The curve supports a collapse diagnosis based on representation/export formulation rather than an obvious optimization divergence.

E. Downstream Protocol and Ablation Scope

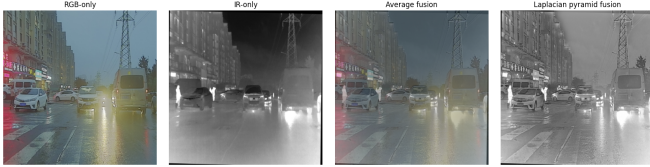
Detection uses YOLO11n with image size 640, batch size 8, and 40 epochs. The verified RGB, IR, and average checkpoints were loaded from prior successful runs, while the proposed YOLO run uses the corrected gate-blend export. Segmentation uses U-Net with a MobileNetV2 encoder, ImageNet initialization, 9 classes, 480×640 input, AdamW at 10^{-4} , batch size at most 4, and 40 epochs.

A3-A6 ablations were trained and evaluated with image-level fusion metrics only. Propagating A3-A6 through downstream YOLO and U-Net would require eight additional downstream trainings and fused-data exports, which were not part of the verified final run. Therefore, A3-A6 support architectural diagnosis at the fusion-metric level only; downstream conclusions are restricted to A0, A1, A2, and A7.

IV. EXPERIMENTAL RESULTS

A. Training and Collapse Diagnostics

The notebook stores 30 CM-JEPA history rows and a checkpointed training curve (Fig. 4). The losses do not indicate simple divergence: the final training total is approximately 0.897, with a JEPA term near 0.037 and a fusion term near 0.839. Validation loss fluctuates, ending near 0.980 with a larger JEPA component. This pattern suggests that the issue is not only insufficient training time. The more precise diagnosis is a formulation and export limitation: the latent objective and raw decoder did not impose enough full-resolution semantic and boundary pressure to make the decoded image reliable. The gate-blend repair produced meaningful images, but it remained a coarse convex mixture rather than a task-aware dense representation.



(a) RGB, IR, average, and raw learned output examples from the notebook.



(b) Corrected gate-blend comparison used for downstream experiments.

Fig. 5: Representative qualitative outputs. The gate-blend repair avoids the collapsed raw decoder image, but it does not guarantee preservation of fine segmentation boundaries.

B. Qualitative Inspection

Qualitative inspection was necessary because the raw proposed decoder could produce nearly uniform gray images. The corrected gate-blend representation preserved road structure, vehicles, lamps, and thermal saliency. However, it also tended to behave like a smoothed, spatially gated mixture. This is visually acceptable for object localization but less appropriate for dense segmentation, where road edges, small structures, and class boundaries must remain sharp.

C. Downstream Perception

Table II and Fig. 6 summarize downstream results. Average fusion is the strongest method on all three task metrics. For

detection, CM-JEPA gate-blend achieved $\text{mAP}@0.5 = 0.7567$ and $\text{mAP}@0.5:0.95 = 0.5080$, close to average fusion’s 0.5115 strict mAP and higher than RGB-only’s 0.4887. Thus, the learned gate preserved useful object-level saliency. For segmentation, CM-JEPA gate-blend achieved only 0.4774 mIoU, below RGB-only, IR-only, and average fusion.

D. Segmentation Error Analysis

The segmentation gap is consistent with the design of the gate-blend representation. The learned gate is produced on a 32×32 feature grid for 512×512 fusion inputs and then upsampled. A single gate cell therefore summarizes a relatively large image region. This is sufficient for broad modality

TABLE II: Downstream performance of RGB, IR, average fusion, and CM-JEPA gate-blend.

Method	mAP@0.5	mAP@0.5:0.95	mIoU
RGB-only	0.7616	0.4887	0.5307
IR-only	0.7064	0.4649	0.4976
Average fusion	0.7718	0.5115	0.5551
CM-JEPA gate-blend	0.7567	0.5080	0.4774

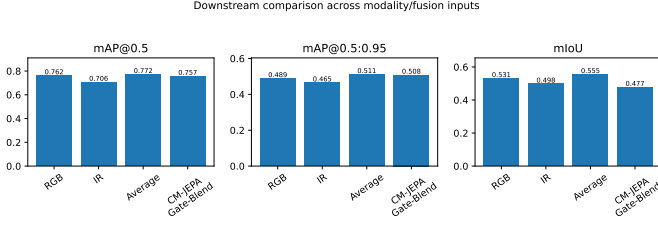


Fig. 6: Downstream results. Average fusion is the strongest overall representation; CM-JEPA gate-blend is competitive in strict detection localization but weak for semantic segmentation.

selection around vehicles or lamps, but it can blur exactly the boundaries that semantic segmentation requires, such as road-sidewalk, vehicle-road, vegetation-building, and sky-building transitions. Average fusion, although simple, preserves full-resolution RGB texture and IR intensity without imposing a learned low-resolution switching field. This explains why CM-JEPA remains close for detection but loses substantially in mIoU.

E. Ablation and Fusion Metrics

Table III reports image-level metrics. The repaired A7 row is central: after replacing the raw decoder metric computation with the same gate-blend representation used downstream, A7 becomes close to average fusion in EN, SF, AG, and NMI. This confirms that the corrected representation is valid at the image-statistics level. It also exposes the limitation of those metrics: A7 and average fusion look nearly identical by EN/SF/AG/NMI, yet their segmentation mIoU differs by 0.0777.

F. Metric-Task Correlation

The correlation analysis uses only the four methods with both fusion metrics and downstream task metrics (A0, A1, A2, A7). It is therefore descriptive rather than statistically conclusive. The updated Pearson report after the A7 repair shows large p-values and unstable relationships. The qualitative message is nevertheless useful: image-level fusion statistics are weak proxies for task utility. High entropy does not imply high detection or segmentation performance, and near-identical EN/SF/AG/NMI does not imply near-identical mIoU.

V. DISCUSSION

A. Main Interpretation

The strongest empirical conclusion is that RGB/IR fusion is useful, but not all fusion is equally useful. Average fusion consistently improves over RGB-only and IR-only in both downstream tasks. This is scientifically important because

Pearson correlation: fusion metrics vs. task metrics (n=4)

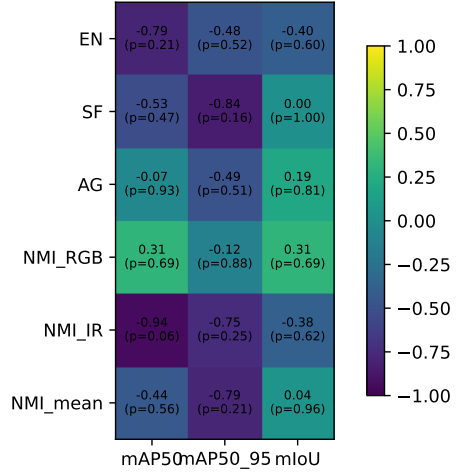


Fig. 7: Descriptive Pearson correlation report after the A7 repair. With only four downstream-evaluated methods, the analysis should not be interpreted as a statistical test.

it establishes that complementary visible and thermal information helped in the lightweight setting. At the same time, the learned CM-JEPA gate-blend model did not improve the overall result. Its advantage is narrower: it is close to average fusion for strict detection localization and improves over RGB-only on mAP@0.5:0.95, but it does not preserve dense semantic information well enough for segmentation.

B. Relation to Learned SOTA Fusion

Compared with DenseFuse and CDDFuse, this project uses a simpler fused-image adapter and a smaller downstream-oriented protocol. Compared with SeAFusion, TarDAL, and MRFS, it does not train the fusion representation jointly with semantic or detection losses. That difference is likely decisive. Task-aware SOTA methods can receive direct gradients from segmentation or detection objectives, whereas the present CM-JEPA module is mainly trained through unsupervised fusion and latent prediction losses. Therefore, the result should not be read as evidence against learned IVIF in general. It is evidence that a task-agnostic gate-blend JEPA module is insufficient to beat a strong average-fusion baseline for both detection and segmentation.

C. Training Budget or Formulation Issue

The collapse diagnosis is primarily a formulation issue, with training budget as a secondary factor. Additional epochs might improve the learned gate or decoder, but they would not remove the central mismatch: the raw decoder is optimized without downstream semantic supervision, and the final gate operates at a coarse spatial resolution. The corrected gate-blend export is a reasonable repair because it uses a stable convex combination of the original modalities, but it also restricts the representation family. Future work should therefore emphasize task-aware objectives, full-resolution or edge-aware gates, bounded residual fusion, and explicit logging of final gate mean, variance, entropy, and saturation statistics.

TABLE III: Fusion-quality and ablation metrics. A3-A6 are fusion-only ablations; downstream metrics were measured only for A0, A1, A2, and A7.

Method	Description	EN	SF	AG	NMI mean
A0	RGB-only	6.2097	0.04021	0.01517	1.5336
A1	IR-only	7.6092	0.03879	0.01275	1.4436
A2	Average fusion	6.8732	0.02836	0.01141	1.1617
A3	Concat, no gate	7.3225	0.01873	0.01022	1.0744
A4	No channel attention	7.3275	0.01898	0.01038	1.0715
A5	No spatial gate	7.2397	0.01715	0.00916	1.0805
A6	Full attention, no JEPA	7.2156	0.01826	0.00976	1.0754
A7	Full CM-JEPA gate-blend	6.8898	0.02834	0.01138	1.1635

D. Limitations and Future Work

The main limitations are clear. A3-A6 ablations were not propagated through full downstream training, so their conclusions remain image-metric based. DenseFuse, CDDFuse, and other learned SOTA methods were not executed under the same YOLO/U-Net protocol, so this work does not claim SOTA comparison. The metric-task correlation uses only four downstream-evaluated methods and should be treated as descriptive. The experiments assume aligned RGB/IR pairs; sensor misalignment would require registration or robustness mechanisms. Future work should evaluate at least one learned SOTA fusion baseline through the same downstream protocol, train all ablation variants through YOLO and U-Net, add segmentation overlay failure cases, and incorporate downstream task losses into the fusion model.

VI. CONCLUSION

This project evaluated CM-JEPA gate-blend RGB/IR fusion for lightweight detection and semantic segmentation. The final result is nuanced: fusion helps, but average fusion is the strongest overall method in the verified experiments. CM-JEPA gate-blend is competitive for object detection under mAP@0.5:0.95, but it underperforms for semantic segmentation because the gate-blend representation does not preserve dense boundary and texture information as effectively as average fusion. The repaired A7 protocol makes the evaluation internally consistent, and the descriptive correlation analysis shows why image-level fusion metrics cannot replace downstream testing. The most defensible next step is not merely to train longer, but to make the fusion model task-aware and to compare it against learned IVIF baselines under the same downstream protocol.

ACKNOWLEDGMENT

The author used AI tools during the preparation of this study for code debugging assistance, experiment interpretation, writing organization, and LaTeX report drafting. The experimental design decisions, final code execution, result selection, and academic interpretation were reviewed and finalized by the author. Responsibility for the submitted content remains entirely with the author.

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